

Tahmid Zaman Tahi

📍 Virginia, USA ✉️ tahi.tahmid@gmail.com 🌐 TahmidZee in [tahmid-zaman-tahi-a16083213](https://www.linkedin.com/in/tahmid-zaman-tahi-a16083213)

Education

- | | |
|--|---------------------------------------|
| Virginia Commonwealth University , Electrical and Computer Engineering | Richmond, Virginia, USA |
| ing | 2025 – present |
| <ul style="list-style-type: none">• Graduate Research Assistant advised by Dr. Yanxiao Zhao.• Research focus on intelligent reflective surface-assisted wireless communications.• Interests include wireless localization, signal processing, machine learning, and connected systems. | |
| The George Washington University , Electrical Engineering | Washington, District of Columbia, USA |
| | 2023 – 2024 |
| <ul style="list-style-type: none">• Graduate Fellowship Award.• Coursework included stochastic processes, machine intelligence, reinforcement learning, wireless networks, communication theory, and digital signal processing. | |
| Islamic University of Technology , Electrical and Electronic Engineering | Gazipur, Bangladesh |
| | 2017 – 2021 |
| <ul style="list-style-type: none">• Undergraduate thesis on low-latency HARQ retransmission for C-V2X sidelink communication.• OIC Scholarship recipient. | |

Selected Publications and Manuscripts

Context-Aware Functional Security for V2X-Enabled Connected Vehicles

Tahmid Zaman Tahi, Sujay Saha, Mostafa Zaman, Yanxiao Zhao, Sherif Abdelwahed
epapers2.org/itec2026/ESR/session_view.php?print=1&session_id=31 🔗

Deep Learning-Enabled Energy-Efficient Joint CSI Estimation and Phase Shift Prediction in Large-Scale RIS-Assisted NOMA Systems (Accepted at ICCCN 2026)

Syed Samiul Alam, Yanxiao Zhao, Haolin Tang, Tahmid Tahi, Wei Wang, Nibir K. Dhar, Min Song
www.icccn.org 🔗

BR-FiLM: Bounded Residual Channel-Quality Conditioning for Automatic Modulation Recognition (Manuscript under review at MILCOM 2026)

Tahmid Zaman Tahi, Syed Samiul Alam, Haolin Tang, Yanxiao Zhao, Jun Huang, Min Song

Enhancing IRS Localization via Deep Learning-Based AOA and Distance Estimation

Nikhila Kotlapara Surendran, Syed Samiul Alam, Haolin Tang, Tahmid Tahi, Wei Wang, Yanxiao Zhao
doi.org/10.31875/2979-1081.2025.01.03 🔗

Resource Allocation in C-V2X: A Review

Tahmid Zaman Tahi
arxiv.org/abs/2401.15756 🔗

HARQ Retransmission Together With Original Transmission on the Same Subframe for C-V2X Communication

K. M. A. Yeaser, M. A. H. Emon, Tahmid Zaman Tahi, I. R. Aninda

Skills

Wireless Communications and Physical Layer

- Multi-antenna and MIMO systems, antenna arrays and steering vectors, beamforming and phase control, CSI and channel estimation, large-scale RIS/IRS, NOMA and successive interference cancellation.
- Digital modulation and detection, link budgets, path loss and fading, SNR/SINR/BER analysis, throughput, spectral and energy efficiency, C-V2X/NR-V2X, HARQ, and resource allocation.

Array Signal Processing and Wireless Sensing

- Complex baseband and I/Q processing, sample and learned covariance, MUSIC, MVDR/Capon, spatial spectra, AoA/range estimation, near-field and multi-source localization.
- Geometric multipath and cascaded RIS channels, block fading and mobility models, partial/full CSI reconstruction, automatic modulation recognition, and low-SNR robustness.

Machine Learning for Communications

- PyTorch, TensorFlow, scikit-learn, CNN/RNN/LSTM models, transformers, FiLM and conditional networks, diffusion models, reinforcement learning, deep Q-learning, multi-objective learning, representation learning, and hyperparameter optimization.

Scientific Computing and Experimentation

- Python, MATLAB, Simulink, NumPy, SciPy, pandas, Matplotlib, C, shell scripting, Git, LaTeX, GPU experiments, and reproducible research pipelines.

Networking and Systems

- Cellular networking and planning, 5G NR-V2X/C-V2X, latency and reliability, resource allocation, IoT/NMS, connected-vehicle security, context-aware anomaly analysis, Atoll, Cisco Packet Tracer, Linux, and user acceptance testing.

Research Engineering Tools

- GitHub, LaTeX, HTML/CSS, Node.js, PSpice, Proteus, PSIM, ARM Keil, and JMP.

Awards

Graduate Fellowship Award

The George Washington University

OIC Scholarship

Islamic University of Technology

Higher Secondary Board Scholarship

Dhaka Education Board

Experience

Virginia Commonwealth University, Graduate Research Assistant

Jan 2025 – present

- Conducting research on intelligent reflective surface-assisted wireless communications with Dr. Yanxiao Zhao.
- Developing learning-enabled signal-processing methods for wireless localization and communication systems.

The George Washington University, Graduate Teaching Assistant

Jan 2023 – May 2024

- Led recitations and supported courses in circuits, signals and systems, digital signal processing, engineering analysis, and computer organization.
- Assessed coding and mathematical work and provided technical support to approximately 60 students per semester.

ADN Telecom Limited, Engineer, Network Infrastructure

Nov 2021 – Dec 2022

- Developed link budgets, analyzed SNR, and performed network planning, surveys, troubleshooting, and technical documentation.
- Supported an IoT-based network-management deployment monitoring power devices across 41 universities.
- Conducted user acceptance testing and collaborated on network-infrastructure delivery.